STATISTICS PRACTICE PAPER

AS-LEVEL

The total for this paper is 60 marks

Advice:

- **Read every question carefully, highlight/underline key information.**
- **Plan a route to the answer, don't mindlessly calculate, as you might miss something**
- **Make sure your answer is reasonable, can a cat be 50 meters tall?**
- **Attempt every question, the first few marks might not be as hard to achieve**
- **When you check a question, assume you made a mistake, then try to find it**
- **Finish the paper quickly, but accurately, and then quadruple check it. Your second set of eyes is always better than your first**
- **Be clear with your method, minimise the number of arrows or boxes you use.**

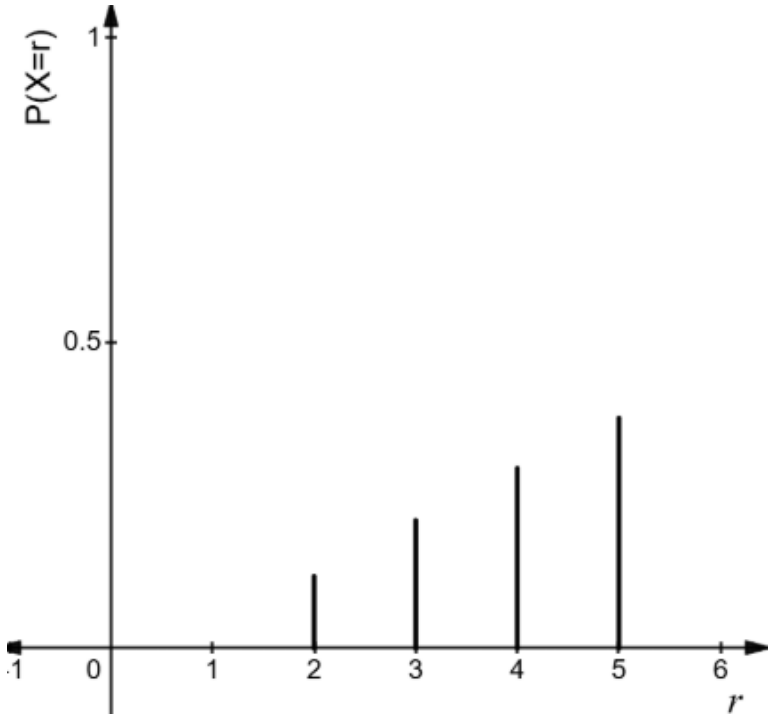
Mark totals:

1 – 7 marks, 2 – 10 marks, 3 – 8 marks, 4 – 11 marks, 5 – 11 marks, 6 – 13 marks

1 A Discrete Random Variable X has the following distribution

$P(X = r) = k\left(r - \frac{1}{r}\right)$ For $r = 2,3,4,5$ and 0 for all other values of r

Provided is a sketch of the distribution



(i) (A) Describe the distribution **[1]**

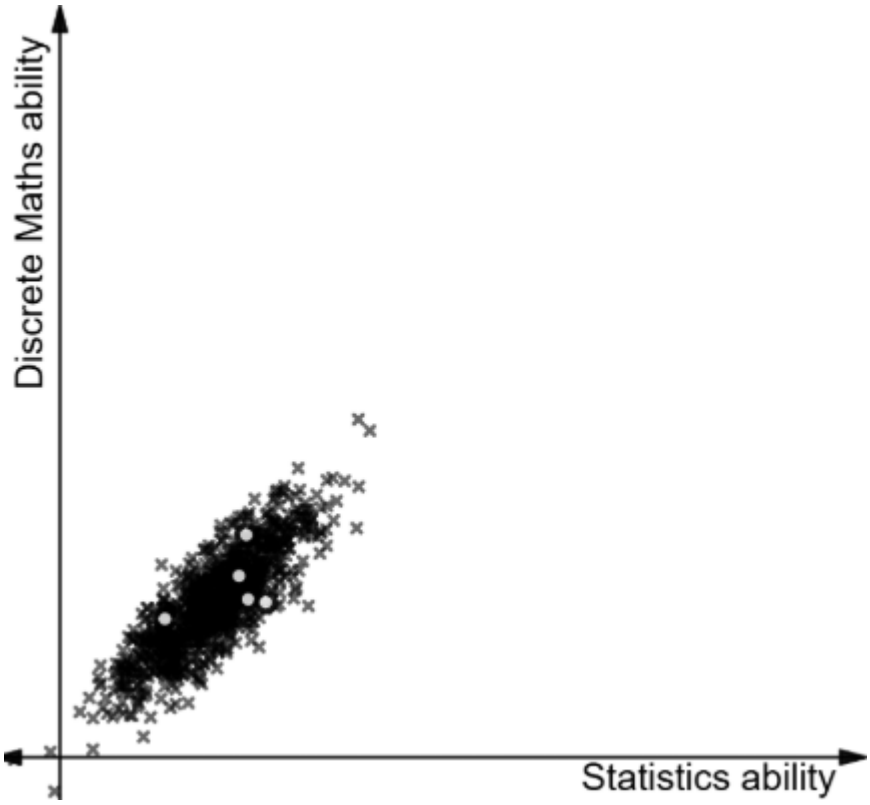
(B) Explain why the distribution has this shape **[1]**

(ii) Show that $k = \frac{60}{763}$ **[2]**

(iii) Calculate each of the following **[3]**

- $\mathrm{E}(X)$
- $\mathrm{Var}(X)$

2 A student is under the belief that there is a relationship between statistics ability and discrete maths ability of further maths students within their city.
To test this, they develop a continuous rating system that allows them to measure each quantifiably.
A graph of data collected from the population of 1000 further maths students is provided below



(i) Explain why carrying out a test based on the PMCC is suitable for the data provided. **[2]**

The student then collects a sample of 30 students from the population
They calculate a PMCC of 0.511

(ii) Carry out a test at the 5% significance level to determine whether there exists any correlation between a further maths student's statistics ability and discrete maths ability **[5]**

(iii) By considering the sample size, explain why the null hypothesis is still likely to be rejected at the 2.5% significance level **[1]**

3 The number of fish sold by a fish shop per minute can be modelled using a binomial distribution, where n represents the number of seconds in a minute.

$$B(60,0.0027)$$

(i) (A) Explain why a Poisson distribution may also be a suitable model **[1]**

(B) State the additional modelling assumptions required for a Poisson distribution to be a suitable model **[2]**

(ii) (A) State the expected number of fish sales per day (N), given that the shop is open for 540 minutes per day **[1]**

(B) Calculate the probability that more than 90 fish is sold in one day **[1]**

The number of bowls of chips sold per day can be modelled by the following Poisson distribution $Y \sim \text{Po}(78)$

(iii) (A) Calculate the probability that exactly 70 bowls of chips and exactly 80 fish are sold in one day **[2]**

(B) Calculate $P(\mu - \sigma \leq N + Y \leq \mu + \sigma)$ **[3]**

4 SCE decides to measure the number of times an old dog barks up the wrong tree within a single day, over the course of 100 days

An article online suggests that the Binomial model $X \sim B(300,0.012)$ is a good model for the number of times this occurs over 300 observations made within a day

(i) State the modelling assumptions required for the model to be suitable **[2]**

SCE want to test if a Poisson distribution with a mean equal to the binomial's follows their own observations.

Provided are their own observations.

Observed occurrences per day	≤ 1	2	3	4	5	$6 \geq$
Frequency	12	18	21	18	14	17

(ii) Explain why the columns 0 and 1 have been merged into ≤ 1. **[1]**

(iii) Carry out a χ^2 test at the 5% significance level, to determine whether the Poisson model is a good fit for the data collected **[8]**

5 An A level maths student decides investigate the relationship between height above sea level h and air density d.

They collect measurements of air density at height intervals of 250m, starting at 0m and ending at 2000m (inclusive of both).

Below are their summary statistics

$$\sum h^2 = 1.275 \times 10^7 \ \sum h = 9000 \ \sum d^2 = 9.54 \ \sum d = 9.16$$

$$\sum hd = 7930$$

(i) Explain why a h on d regression line is not suitable to calculate **[1]**

(ii) Predict the density of air at the following heights above sea level **[4]**

- 490m
- 2700m

(iii) Make a comment on the reliability of your predictions in part (ii) **[2]**

(iv) Calculate the residual for the point $(1750, 1.01)$ **[2]**

The student claims

"A change in height above sea strongly implies a change in air density"

The value of the PMCC for the sample is -0.996

(v) Make a comment on the students claim. **[2]**

6 Let X be a random variable uniformly distributed across the distinct values $\{2,9,\ldots,7n-12,7n-5\}$ in a fair electronic spinner.

(i) (A) In the event n is even, State $P\left(X < \frac{7n-3}{2}\right)$ **[1]**

(B) In the event n is odd, calculate $P\left(X < \frac{7n-3}{2}\right)$
Leaving your answer as an algebraic fraction **[3]**

(ii) Calculate each of the following, Leaving your answers as algebraic fractions **[4]**

- $\mathrm{Var}(X)$
- $\mathrm{E}(X)$

A student then rolls the spinner until the first time they roll a 9.

(iii) (A) By letting N represent the number of spins until the first 9,
form a model using the geometric distribution to represent this information,
in terms of n. **[1]**

(B) Given that $P(N \geq 3) = \frac{81}{100}$
Calculate the value of n **[3]**

(C) State the variance of the number of spins until the first 9 **[1]**